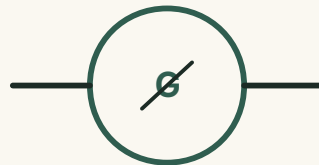


# Voltmeters & Ammeters

How one sensitive movement becomes two measuring instruments.

## ESSENTIAL QUESTION

**Why does a voltmeter need a large resistance, while an ammeter needs a small one?**



sensitive movement

● ROUTE

# The measurement goal determines the structure.

01

## LIMIT

A movement tolerates only a small full-scale current.

02

## VOLTAGE

Add series resistance to limit current.

03

## CURRENT

Add a parallel shunt to bypass current.

04

## RANGE

Select resistance to set the full-scale value.

### THE CONSTRAINT

The circuit must **protect the movement** at full scale.

## MODEL

# A galvanometer is sensitive — and limited.

Its movement reaches full-scale deflection at a small current  $I_g$ .

$I_g$  full-scale current

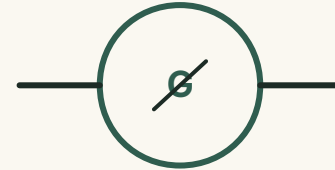
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$R_g$  internal resistance

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$U_g$  voltage at full scale,  $U_g = I_g R_g$

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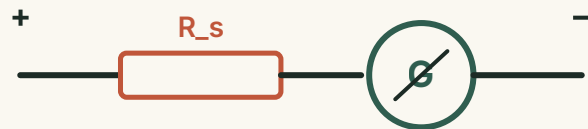
sensitive movement

## CONVERT

# A voltmeter adds resistance in series.

The multiplier resistor takes most of the applied voltage.

- 1 Limit the current**  
Series resistance keeps the movement current at  $I_g$ .
- 2 Large input resistance**  
So the meter barely disturbs the circuit.
- 3 Connect in parallel**  
The finished voltmeter goes across the two points.



series multiplier · connect in parallel

## DERIVE

# The series resistor sets the voltage range.

At the chosen full-scale voltage  $V$ , the movement current must equal  $I_g$ .

VOLTAGE ACROSS THE CHAIN

$$V = I_g(R_s + R_g)$$

SOLVE FOR THE MULTIPLIER

$$R_s = \frac{V}{I_g} - R_g$$

RANGE MULTIPLIER  $N = V / U_g$

$$R_s = (n - 1)R_g, \quad R_V = nR_g$$

Larger voltage range  $\rightarrow$  larger series resistance.

## CONVERT

# An ammeter adds a low-resistance shunt.

The movement carries  $I_g$ ; the shunt carries the rest.

### 1 Split the current

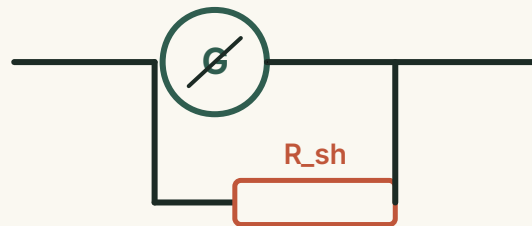
At full scale,  $I = I_g + I_{sh}$ .

### 2 Same voltage

Both branches share the same voltage.

### 3 Connect in series

The finished ammeter goes in the line.



parallel shunt • connect in series

## DERIVE

# The shunt sets the current range.

Parallel branches share the same voltage.

EQUAL BRANCH VOLTAGES

$$I_g R_g = (I - I_g) R_{sh}$$

SOLVE FOR THE SHUNT

$$R_{sh} = \frac{I_g R_g}{I - I_g}$$

RANGE MULTIPLIER  $N = I / I_g$

$$R_{sh} = \frac{R_g}{n - 1}, \quad R_A = \frac{R_g}{n}$$

Larger current range  $\rightarrow$  smaller shunt resistance.

## COMPARE

# The two conversions have opposite design goals.

Series limits current; parallel redirects current.

### VOLTMETER

Series multiplier · large internal resistance ·  
connect in **parallel** with the circuit.

### AMMETER

Parallel shunt · small internal resistance ·  
connect in **series** with the circuit.



## EXTEND

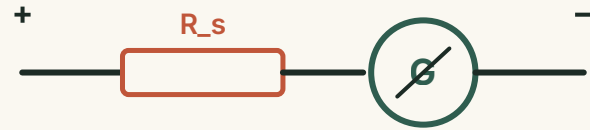
# A multi-range voltmeter selects series resistance.

Higher-voltage terminals include more resistance in the current path.

### RANGE RULE

Higher range → more series resistance.

Always start on the highest range.



series multiplier · connect in parallel

## EXTEND

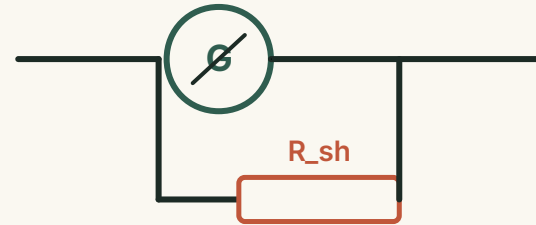
# A multi-range ammeter selects a shunt.

The chosen shunt controls how much current bypasses the movement.

### RANGE RULE

**Higher range → smaller shunt resistance.**

Never connect an ammeter directly across a supply.



parallel shunt · connect in series

## APPLY

# Worked example: design two voltage ranges.

Movement:  $I_g = 100 \mu\text{A}$ ,  $R_g = 500 \Omega$ . Design 3 V and 15 V ranges.

### 1 3 V range

$$R_1 = \frac{3}{100 \mu\text{A}} - 500 = 29.5 \text{ k}\Omega$$

### 2 15 V range (in series with $R_1$ )

$$R_2 = 120 \text{ k}\Omega$$

#### CHECK

Both ranges still drive exactly  $I_g$  at full scale.

## APPLY

# Worked example: design a current shunt.

Movement:  $I_g = 10 \text{ mA}$ ,  $R_g = 50 \Omega$ . Design a 1.0 A range.

1

**Apply the shunt formula**

$$R_{sh} = \frac{0.010 \times 50}{1.0 - 0.010} \approx 0.505 \Omega$$

### INTERPRETATION

About 99% of the current bypasses the movement through the tiny shunt.

## CONNECT

# Correct connection protects the circuit and the meter.

Real meters also change the circuit they measure.

### VOLTMETER

Parallel · high R. A finite current still flows, causing slight loading.

### AMMETER

Series · low R. A small voltage drop causes loading. Wrong connection can overload the movement.

- **FINISH**

# The circuit structure explains the instrument.

Protect the movement, choose the range, then connect correctly.

## **VOLTMETER**

Series  $R_s$  · large  $R_V$  · connect in parallel.

## **AMMETER**

Parallel  $R_{sh}$  · small  $R_A$  · connect in series.